Warehouse Slotting Optimizer

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Warehouse Slotting Optimizer** presents a data-driven system for assigning products to suitable storage locations in a warehouse to minimize picking distance, reduce handling time, and improve space utilization. The system analyzes product information such as demand frequency, product size, weight, and storage requirements to determine efficient slot assignments. Frequently picked products can be positioned closer to dispatch and high-traffic areas, while storage constraints are considered to ensure practical placement. The system applies appropriate data structures and optimization algorithms to efficiently manage products, storage slots, and picking requirements. It also evaluates the quality of the generated slotting plan using measures such as total picking distance, estimated retrieval time, space utilization, and workload distribution. The project demonstrates how Data Structures and Algorithms can be applied to solve a real-world warehouse management problem and improve operational efficiency.

Signature of the Guide